

Sec A	(Attempt any 2 questions) (6 x 2)	Sec B	(Attempt any 1 question) (8x1)
1. Find the rank of the matrix by reducing it to normal form	$\begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix}$	1. If $x = \sin t$ and $y = \cos pt$. Prove that	$(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 - p^2)y_n = 0$
2. Show that the matrix B^0AB is Hermitian or skew-Hermitian according as A is Hermitian or skew-Hermitian		2. Test for consistency and solve :	$x + 2y + 2z = 1, 2x + y + z = 2, 3x + 2y + 2z = 3$
3. Find the radius of curvature of the curve $x = a[\cos t + \log \tan(t/2)]$ and $y = a \sin t$.			$y + z = 0$
4. Evaluate $\lim_{x \rightarrow 0} \left[\frac{1}{x} - \cot x \right]$.			

$$\begin{aligned} \sin(A+B) &= \sin A \cos B + \cos A \sin B \\ \sin(A-B) &= \sin A \cos B - \cos A \sin B \\ \cos(A+B) &= \cos A \cos B - \sin A \sin B \\ \cos(A-B) &= \cos A \cos B + \sin A \sin B \end{aligned}$$

Sec A (Attempt any 2 questions) (6 x 2)	Sec B (Attempt any 1 question) (8x1)
1. If $u = \tan^{-1} \left(\frac{x^2+y^2}{x-y} \right)$, prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$. 2. Verify whether $u = \sin^{-1}x + \sin^{-1}y$ and $v = x\sqrt{1-y^2} + y\sqrt{1-x^2}$ are functionally dependent, and if so find the relation between them. 3. Evaluate by changing the order of integration $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$. 4. Evaluate $\int_0^1 x^{n-1} \left[\log \left(\frac{1}{x} \right) \right]^{m-1} dx$.	1. A function $f(x,y)$ is rewritten in terms of new variables $u = e^x \cos y$, $v = e^x \sin y$, prove that $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = (u^2 + v^2) \left(\frac{\partial^2 f}{\partial u^2} + \frac{\partial^2 f}{\partial v^2} \right)$. 2. Evaluate $\iiint_S xyz \, dx dy dz$, where $S = [(x,y,z): x^2 + y^2 + z^2 \leq 1, x \geq 0, y \geq 0, z \geq 0]$